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COMPUTER ENGINEERING DEPARTMENT

CENG 408

Discussion, Experiments & Results
Report

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1. Introduction to the Discussion

This chapter discusses the MedVisVR project: experimental results, methodology, limitations, ethical considerations, scalability, and comparisons with related work. We try to place the numbers in context and be explicit about where the system is still weak. One constraint is worth stating up front: MedVisVR runs entirely offline. Every stage of the pipeline, from segmentation to report drafting to VR rendering, executes on local hardware with no outbound network traffic. The motivation is patient privacy and usability in environments where cloud services are not a safe default.

2. Experiments and Results

This section presents the experimental setup, training and validation configurations, evaluation metrics, and key results obtained from the brain tumor, lung nodule, and liver tumor segmentation pipelines. Initial baseline experiments were conducted during the CENG 407 phase of the project, and these were subsequently extended with refined configurations and additional evaluations in the CENG 408 phase.

2.1 Training and Validation Setup

2.1.1 Brain Tumor Segmentation (nnU-Net v2 – BraTS 2020)

The brain tumor segmentation model was trained using the nnU-Net v2 framework on the BraTS 2020 dataset. Training follows nnU-Net’s self-configuring approach, where the framework automatically selects the preprocessing pipeline, network architecture, and hyperparameters based on the dataset.

Dataset Split:

Split	Number of Cases	Purpose
Training	369	Model training with 5-fold CV
Validation	125	Hyperparameter tuning & model selection

Training Configuration:

Parameter	Value
Architecture	3D U-Net (nnU-Net v2 auto-configured)
Input Modalities	T1, T1ce, T2, FLAIR (4-channel)
Patch Size	$128 \times 128 \times 128$
Voxel Spacing	$1.0 \times 1.0 \times 1.0$ mm (isotropic)
Normalization	Z-score (per-modality, brain region only)
Loss Function	Dice Loss + Cross-Entropy (compound)

Optimizer	SGD with Nesterov momentum
Learning Rate Schedule	Polynomial decay
Cross-Validation	5-fold
Data Augmentation	Random flips, rotations, scaling
Class Balancing	Foreground oversampling (2:1 positive:negative ratio, nnU-Net default)
Determinism	seed=0 for reproducibility

We restructured the original BraTS labels (0, 1, 2, 4) into three clinically meaningful channels: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET), which is standard practice for Dice-based BraTS evaluation.

2.1.2 Lung Nodule Segmentation (nnU-Net v1 – Task06_Lung)

The lung CT segmentation pipeline uses nnU-Net v1 (3d_fullres) with the Task06_Lung pretrained weights from the Medical Segmentation Decathlon (Simpson et al., 2019). No retraining was performed. Inference uses the 5-fold ensemble with test-time augmentation. Quantitative evaluation in this report is performed on the fold_0 validation split (13 cases).

Dataset Characteristics:

Parameter	Value
Source	Medical Segmentation Decathlon, Task06_Lung
Total Cases	63 chest CT volumes
Annotation	Binary voxel-level (label=1: nodule)
Image Dimensions	512 × 512 × N (N varies per case)
Format	NIfTI (.nii.gz)
Inference Mode	5-fold ensemble with test-time augmentation

2.1.3 Liver Tumor Segmentation (nnU-Net v1 – Task003_Liver)

The liver tumor segmentation pipeline uses nnU-Net v1 (3d_fullres) with the Task003_Liver pretrained weights from the Medical Segmentation Decathlon (Simpson et al., 2019). No retraining was performed. The full 5-fold ensemble is applied at inference time. Quantitative evaluation is performed on a 10-case held-out cohort (Dataset603_Liver).

The pipeline distinguishes three semantic labels: 0 (background), 1 (liver parenchyma), and 2 (liver tumor). This allows the downstream metric extraction layer to separate total liver volume from tumor burden and to report them independently. Inputs are expected as abdominal CT volumes in NIfTI format following the nnU-Net naming convention with the trailing _0000.nii.gz modality suffix, and segmentation is launched through the nnUNet_predict entry point with

RESULTS_FOLDER pointing to the local pretrained weight store, keeping the liver stage fully offline and on the same infrastructure as the brain and lung stages.

Dataset Characteristics:

Parameter	Value
Source	Medical Segmentation Decathlon, Task003_Liver (pretrained weights)
Evaluation cohort	10 abdominal CT cases (Dataset603_Liver held-out)
Labels	0: background, 1: liver parenchyma, 2: liver tumor
Format	NIfTI (.nii.gz)
Inference Mode	5-fold ensemble (folds 0–4)

2.2 Evaluation Metrics

We evaluated segmentation performance across all three pipelines using the following metrics:

Metric	Formula	Purpose
Dice Similarity Coefficient (DSC)	$2 \cdot TP / (2 \cdot TP + FP + FN)$	Overall segmentation overlap accuracy
Precision	$TP / (TP + FP)$	False positive control
Recall (Sensitivity)	$TP / (TP + FN)$	Lesion detection completeness
Specificity	$TN / (TN + FP)$	True-negative control under extreme class imbalance
Balanced Accuracy / AUC	$\frac{1}{2} (TPR + TNR)$ (equals AUC for hard predictions)	Class-balanced accuracy, robust to imbalance
F1 Score	$2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall})$	Equals Dice for binary labels; reported explicitly for consistency
AUC	Area under the ROC curve (equals Bal. Acc. on hard predictions)	Discrimination between foreground and background
Macro F1	Unweighted mean of per-class F1 (incl. background class)	Multi-class aggregate for brain (NCR/Edema/ET) and liver (Parenchyma/Tumor)

Per-case metrics are computed independently and then macro-averaged across cases. For binary foreground labels Dice equals F1 and Balanced Accuracy equals AUC on hard predictions, so these are reported together in the result tables.

2.3 Brain Tumor Segmentation Results

Per-case quantitative inference was run on the fold 0 held-out subset ($n = 5$), for which ground-truth labels were locally available. The official BraTS 2020 validation split ($n = 125$) listed in 2.1.1 is provided for dataset completeness; its ground-truth labels are not publicly released by the

challenge organizers, so it is not used for quantitative inference in this report. The 5-fold cross-validation EMA Dice from the nnU-Net training checkpoint (reported below) provides a cohort-level cross-check over the full 369-case training data.

Per-label metrics are reported for the three raw BraTS label classes (NCR/lbl 1, peritumoral edema/lbl 2, enhancing tumor/lbl 4). The standard composite sub-regions are Whole Tumor (WT = lbl 1+2+4), Tumor Core (TC = lbl 1+4), and Enhancing Tumor (ET = lbl 4). These are derived from the labels as described in §2.1.1 and used for summary reporting.

Table: Brain Tumor Segmentation Performance by Sub-Region

The nnU-Net v2 fold 0 inference on the BraTS 2020 validation split ($n = 5$ cases) produced the following per-label scores. These five cases represent the fold 0 held-out split used for evaluation. Labels follow the BraTS schema where label 1 is non-enhancing/necrotic core (NCR), label 2 is peritumoral edema, and label 4 is the enhancing tumor (ET).

Sub-region	Dice	Precision	Recall	Specificity	Bal. Acc.	F1	AUC	Macro F1
NCR / Non-enh. (lbl 1)	0.801 $\pm$ 0.117	0.906 $\pm$ 0.019	0.736 $\pm$ 0.182	0.965	0.868 $\pm$ 0.061	0.801 $\pm$ 0.117	0.861 $\pm$ 0.091	0.873 $\pm$ 0.057
Edema (lbl 2)	0.825 $\pm$ 0.115	0.883 $\pm$ 0.078	0.796 $\pm$ 0.168	0.981	0.898 $\pm$ 0.084	0.825 $\pm$ 0.115	0.891 $\pm$ 0.082	—
Enhancing Tumor (lbl 4)	0.868 $\pm$ 0.023	0.868 $\pm$ 0.038	0.869 $\pm$ 0.024	0.992	0.932 $\pm$ 0.013	0.868 $\pm$ 0.023	0.974 $\pm$ 0.012	—

Table: Per-Fold Cross-Validation Results

Rather than regenerating per-fold tables, we report the 5-fold EMA Dice from the nnU-Net training checkpoint as a compact cross-validation summary.

Source	Dice (mean $\pm$ std)
Fold 0 validation inference, $n = 5$ (per-label averages above)	0.80–0.87 per sub-region
5-fold CV EMA Dice from checkpoint (all training data)	0.8691 $\pm$ 0.0019

Per-sub-region Dice sits in the 0.80–0.87 range with macro-F1 0.873 $\pm$ 0.057, and the 5-fold checkpoint EMA Dice is 0.8691 $\pm$ 0.0019. These results are consistent with published BraTS 2020 nnU-Net baselines and fall short of top-leaderboard ensembles, which rely on larger model stacks and heavier post-processing. That gap is expected here, because segmentation is used as a pretrained backbone for downstream visualization and reporting rather than as a leaderboard submission. The small EMA standard deviation (0.0019) reflects exponential moving-average smoothing within the training checkpoint rather than cross-fold variance. Note on Macro F1: it is a single global value (0.873 $\pm$ 0.057) computed across all label classes (background, NCR, edema, ET), not a per-sub-region statistic, so the cohort value (0.873 $\pm$ 0.057)

is shown in the first data row; the other rows show — to indicate the column is not applicable per-label.

2.4 Lung Nodule Segmentation Results

The lung nodule segmentation was evaluated on the fold 0 validation split of the Medical Segmentation Decathlon Task06_Lung dataset ($n = 13$ cases; the pretrained model was trained for 250 epochs by the Decathlon benchmark authors (Simpson et al., 2019)—no retraining was performed in this work, consistent with §2.1.2). The case-level Dice distribution is bimodal: most cases cluster above 0.78, while a small number produce very low or zero overlap, including one case (lung_033) where the model predicted only 115 voxels against 39,103 ground-truth voxels. Specificity is essentially saturated at 0.99994 because of the extreme class imbalance between nodule and non-nodule voxels, so Dice, precision and recall are the informative metrics.

Table: Lung Nodule Segmentation, fold 0 Validation Set ($n = 13$)

Cohort	Dice	Precision	Recall	Specificity	Bal. Acc.	F1	AUC	Macro F1
Lung CT (fold_0 val, $n=13$)	0.652 ± 0.271	0.641 ± 0.315	0.717 ± 0.259	0.938	0.859 ± 0.129	0.652 ± 0.271	0.859 ± 0.129	N/A

Table: Notable Cases (fold 0 validation set)

Case	Dice	Note
lung_046	0.893	Best case (mean $\pm$ std basis)
lung_034	0.879	High precision & recall
lung_006	0.866	Precision 0.918, recall 0.819
lung_033	0.779	Complete under-segmentation failure (115 pred vs 39,103 GT voxels)
lung_079	0.417	Lowest non-zero; high recall but very low precision

Metrics were computed using standard scikit-learn binary segmentation evaluation. We include the lung_033 zero-Dice case rather than dropping it; it is the dominant contributor to the 0.271 Dice standard deviation and its under-segmentation behavior is discussed in §3.2.

2.5 Liver Tumor Segmentation Results

5-Fold Ensemble Evaluation on 10 Liver Cases

Each case is processed through the full 5-fold Task003_Liver ensemble. The pipeline produces a multi-label NIfTI mask (parenchyma + tumor) plus derived metrics (total tumor volume, per-lesion volumes, lesion count) consumed by the descriptor and report layers in the same way as the brain and lung outputs.

A quantitative 5-fold-ensemble evaluation has now been performed on a 10-case liver cohort (Dataset603_Liver). The large tumor-Dice standard deviation is driven by two tumor-free cases (liver_105, liver_106) where the Dice denominator is undefined (0/0) and is assigned 0.0 following standard challenge evaluation conventions (lower-bound reporting), and by one small-tumor case where both precision and recall degrade. A larger held-out cohort with expert annotations remains a target for future work.

Table: Liver Segmentation, 5-Fold Ensemble on 10 Cases (Dataset603_Liver)

Label	Dice	Precision	Recall	Specificity	Bal. Acc.	F1	AUC	Macro F1
Parenchyma (lbl 1)	0.955 ± 0.053	0.937 ± 0.083	0.976 ± 0.015	0.966 ± 0.002	0.988 ± 0.008	0.955 ± 0.053	0.941 ± 0.007	0.887 ± 0.116
Tumor (lbl 2)	0.706 ± 0.357	0.719 ± 0.364	0.694 ± 0.352	0.949	0.823 ± 0.176	0.706 ± 0.357	0.847 ± 0.167	—

Table: Liver, Ensemble vs. 5-Fold CV Cross-Check

Source	Parenchyma Dice	Tumor Dice
5-fold ensemble inference, held-out cohort (n = 10)	0.955 ± 0.053	0.706 ± 0.357
5-fold CV Dice from postprocessing.json (n = 131)	0.9571	0.6372

2.6 Ablation Studies

Three small, targeted ablation checks were run on the pipeline: one on the segmentation preprocessing path, one on the VR rendering configuration, and one on the AI assistant stack. Each check is limited in scope and is meant to confirm the direction of the current design decisions rather than to exhaustively tune them.

2.6.1 Segmentation Preprocessing Ablation

We compared the nnU-Net v2 default preprocessing against two modified variants on a small brain-MRI subset (single fold, fold 0) to check the direction of each choice.

The nnU-Net v2 auto-configured preprocessing served as the baseline (per-modality z-score normalization with use_mask_for_norm enabled on all four channels, isotropic $1.0 \times 1.0 \times 1.0$ mm resampling, 3D patch size $128 \times 128 \times 128$ and 2:1 foreground-to-background oversampling, read from the plans.json bundled with the trained model). Removing the brain-mask restriction on normalization and forcing anisotropic spacing were evaluated as two separate variants. Both variants produced lower Dice on the enhancing-tumor sub-region than the default, with only small changes on whole-tumor overlap. The result is consistent with the nnU-Net design rationale, and the default configuration is kept for the reported runs.

2.6.2 VR Visualization Ablation

The VR rendering configuration was checked on the Meta Quest 3 by toggling the two mobile-specific settings and the intensity enhancement one at a time, using a representative BraTS case.

The baseline operating point (4× intensity enhancement, multi-scale boundary blending, Fixed Foveated Rendering, 0.65× mobile sampling-rate multiplier (~333 effective ray-marching steps down from 512) and tricubic interpolation disabled on Android) was compared against four single-change variants on the Meta Quest 3. Disabling Fixed Foveated Rendering caused the frame rate to drop below the 72 Hz target on larger BraTS volumes. Running at the full sampling rate (no 0.65× multiplier) also caused the frame rate to drop below the 72 Hz target (a distinct breach, not identical in magnitude). Re-enabling tricubic interpolation on Android recovered some boundary smoothness but increased per-frame texture reads and reintroduced occasional reprojection effects during head motion. Reducing the intensity enhancement towards 1× visibly weakened the perceptual separation of the enhancing-tumor sub-region. The baseline configuration was retained for these reasons.

2.6.3 AI Assistant Ablation

The AI assistant was checked by running the evaluation harness in three configurations: the full stack, the stack with the safety-validation layer bypassed, and a direct-LLM variant without the structured descriptor.

Each layer is designed for a specific role: the descriptor pins numerical fields to a fixed PatientContext, the safety-validation layer filters forbidden diagnosis and treatment-prescriptive patterns, and the 2 % tolerance balances drift sensitivity against rounding robustness. Bypassing the safety-validation layer let occasional treatment-prescriptive phrasing through that the full stack rejects. The direct-LLM variant produced more numerical inconsistencies on questions involving volumes and lesion counts, because those fields were re-derived in free text instead of being read from the PatientContext. On a separate 10-case BraTS evaluation cohort used for AI assistant testing (runs_first10_reportllm_relaxed_v1; distinct from the n = 5 fold-0 evaluation split used in §2.3) the full stack reached a mean factual_accuracy of 0.94, hallucination_rate 0.04, omission_rate 0.05 and safety_compliance 0.94, with cohort-level qa_evidence_coverage 0.92, unsupported_statement_rate 0.03, contradiction_rate 0.01 and required_section_missing_rate 0.02. The full stack therefore remained the configuration of record. The harness outputs and aggregate methodology are described in §3.5.

2.7 Baseline Comparison (CENG 407 vs. CENG 408)

The project spans two academic terms. CENG 407 produced a qualitative feasibility baseline (single-fold brain-MRI nnU-Net v2 rendered on Quest 3 through a manual Unity scene). CENG 408 extended it with quantitative multi-cohort evaluation, atlas-backed reasoning, a local LLM QA module, an automated VR pipeline, and a unified desktop application, all without adding a network dependency. The table below summarises the change at the capability level; numbers are in the preceding subsections.

Table: Evolution of Results Across Project Phases

Aspect	CENG 407 Baseline	CENG 408 Extended
Brain tumor segmentation	nnU-Net v2 on BraTS 2020 (3d_fullres, single fold, fold 0)	nnU-Net v2 on BraTS 2020 with fold 0 validation inference and 5-fold CV EMA Dice reported from the training checkpoint; multi-label WT / TC / ET output
Additional modality	Brain MRI only (glioma, BraTS 2020; no lung or liver pipeline)	Brain MRI + chest CT + abdominal CT; lung nodule via nnU-Net v1 Task06_Lung (5-fold ensemble with TTA, fold_0 validation n=13), liver via nnU-Net v1 Task003_Liver (5-fold ensemble, n=10 held-out)
Brain anatomy reference	Fixed dummy brain prototype; tissue extracted via Otsu thresholding and morphological operations	Patient-space segmentation plus Harvard–Oxford atlas-backed anatomic mapping with registration-quality awareness
Spatial alignment	Center-of-mass rigid translation of the tumor into the dummy brain	Center-of-mass rigid alignment retained, extended with atlas-space localization and quality flags
Quantitative tumor metrics	Not computed (segmentation mask only)	Total burden, max diameter, per-label volumes, lesion list, merged lesion-group count, segmented component count
Clinical reporting	Not implemented	Deterministic structured report with fixed sections (Clinical Context, Findings, Impression, Limitations, Recommendations, Disclaimer) and evidence references
AI assistant / QA	Not implemented	Local llama.cpp runtime with Qwen2.5-7B-Instruct (GGUF); evidence-linked QA with guardrails on diagnosis, treatment and prognosis questions
Evaluation framework	Not implemented	Automated per-case artefacts: quality_report.json, metrics.json, eval_summary.json, error_log.json, comparison.json, run_meta.json
VR visualization	Manual Unity scene setup with UnityVolumeRendering plugin deployed on Meta Quest 3	Automated Unity VR deployment pipeline with performance optimizations for Quest 3 (Fixed Foveated Rendering, 0.65× mobile sampling-rate multiplier (~333 effective steps) with tricubic interpolation disabled on Android, locked GPU clock, stable 72 Hz)
Desktop application	Not implemented	MedVisVR desktop application integrating the case viewer, report draft and AI assistant in a single interface
Network dependency	Segmentation runs locally, but pipeline not explicitly designed for disconnected use	End-to-end fully offline by design: segmentation, metrics, atlas mapping, reporting, LLM inference and VR all execute on local hardware with no cloud or external API calls

Reproducibility	Ad-hoc scripts without run-level fingerprinting	Per-case run_meta.json with code fingerprint, prompt version, LLM policy, question-set hash and artefact paths
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3. Interpretation of Results

3.1 Brain Tumor Segmentation Performance

The nnU-Net v2 inference on the BraTS 2020 fold_0 validation split ($n = 5$) yields per-sub-region Dice in the 0.80–0.87 range with macro-F1 0.873 ± 0.057 , and the 5-fold checkpoint EMA Dice of 0.8691 ± 0.0019 is internally consistent with the validation slice. The enhancing-tumor sub-region (label 4) is the most stable (Dice std 0.023), whereas NCR and edema show larger variance, mostly because of lower recall on a small number of weaker cases. These numbers fall within the regime of reported BraTS 2020 nnU-Net single-fold baselines (approximately 0.85–0.90) but below top-leaderboard ensembles with heavy post-processing. That gap is expected and intentional: segmentation here is a pretrained backbone that feeds the downstream interpretation and VR layers, not a benchmark submission.

3.2 Lung Nodule Segmentation Performance

The fold_0 validation mean Dice of 0.652 ± 0.271 on $n = 13$ cases is driven by bimodality, not by uniform degradation. The majority of cases cluster above Dice 0.78 (median 0.791) while one failure case (lung_033) registers Dice 0.0, a 115-voxel prediction against a 39,103-voxel ground truth (correct localization, but under-segmentation by two orders of magnitude). Reporting only the cohort mean would hide this split and suggest uniform accuracy; the per-case view makes the failure mode visible. Specificity saturates at 0.99994 because of the extreme foreground/background imbalance intrinsic to lung-nodule segmentation, which is why Dice, precision and recall (and not pixel-level accuracy) are the informative metrics here. In practice, any deployment of this system would need automated per-case quality checks rather than relying on a cohort-level average.

3.3 Liver Tumor Segmentation Performance

The 5-fold ensemble on the 10-case Dataset603_Liver cohort produces parenchyma Dice 0.955 ± 0.053 and tumor Dice 0.706 ± 0.357 (all 10 cases, lower-bound: two tumor-free cases assigned Dice = 0.0). Excluding liver_105 and liver_106, the tumor Dice on the 8 tumor-present cases is higher than the all-10 average, but the all-10 figure is reported as the conservative lower-bound. The large standard deviation (0.357) is driven by these two tumor-free cases and by one small-tumor case. The parenchyma number is in the expected nnU-Net range for a large, high-contrast abdominal organ. As an independent sanity check, nnU-Net’s own 5-fold cross-validation on the training data ($n = 131$) gives parenchyma 0.9571 and tumor 0.6372; the held-out ensemble result (0.706) is approximately 0.07 higher than the CV estimate, which is within one pooled standard deviation and is consistent with the small held-out sample size rather than systematic overfitting. The main remaining gap is a larger, expert-annotated tumor cohort that would support a tumor-specific baseline rather than a mixed tumor/non-tumor average.

3.4 VR Visualization and Integration

The VR stack is interpreted not as a rendering engine in isolation but as a perceptual channel through which the segmentation results are received. Two decisions shape this channel. First, a 4× intensity enhancement on the enhancing-tumor sub-region combined with multi-scale Gaussian boundary blending makes the tumor boundary visible in the rendering without implying a sharp anatomical edge, which fits the diffuse infiltration pattern typical of glioblastoma. Second, pairing the volumetric view with a live 2D cross-section and an angle readout gives the user the planar view clinicians routinely read, without leaving the headset. The Quest 3 performance envelope (72 Hz sustained via a 0.65× sampling-rate multiplier, disabled tricubic filtering, Fixed Foveated Rendering and locked GPU clocks) comes with a known trade-off: mild voxel aliasing in low-gradient regions and some risk of under-sampling sub-voxel structures. This is acceptable for educational use, but it should be taken into account when interpreting subtle boundary transitions.

3.5 AI Assistant and Clinical Decision Support

We evaluated the assistant on two dimensions: hallucination control and pipeline integrity. Hallucination control is enforced at three layers. A generation-time guardrails module blocks disallowed diagnosis terms and cross-checks every numerical claim against the PatientContext with a 2 % tolerance. A deterministic report generator rejects drafts that miss required sections or match forbidden treatment-prescriptive patterns. An automated harness classifies each answer on a reference question set as supported, hallucination or omission, and emits aggregate hallucination_rate, omission_rate and safety_compliance values per run. Case-locking at the session layer (history discarded on patient_id/study_id change or 1800 s idle TTL) rules out cross-patient contamination through conversational state. Pipeline integrity is measured by a pytest suite covering schema validation, report generation, QA engine, retrieval, case repository, API endpoints, integration and a golden-case sanity pass. The pipeline behavior tests pass across the board; the only open items are a small set of schema-validation tests affected by a known contract-drift between current tumor_metrics.json field names and legacy required keys. This is a validator-side issue scheduled for the next revision and does not reflect a pipeline regression.

4. Analysis of Limitations

4.1 Segmentation-Dependent Pipeline

A central limitation of MedVisVR is that everything downstream depends on segmentation quality. Quantitative metrics, atlas localization, report generation, QA and VR rendering all inherit the errors of the segmentation mask. If the model misidentifies tumor boundaries, under-segments edema or produces fragmented masks, the rest of the pipeline can remain internally consistent while still being clinically misleading.

This is especially problematic because segmentation errors are often subtle and may not be obvious in the VR environment. A clinician or student viewing a volumetrically rendered tumor has no built-in mechanism to assess whether the rendered boundaries accurately reflect the true

pathology. Future work should add segmentation uncertainty visualization, for example using transparency gradients or color-coded confidence maps overlaid on the rendered volume.

4.2 Rigid Registration Limitations

Center-of-mass rigid alignment preserves tumor morphology but sacrifices anatomical precision. This trade-off is acceptable for educational use but limits the system for surgical planning. Deformable registration (ANTs, SimpleElastix) would improve anatomical fidelity, at a significantly higher compute cost and with some risk of distorting tumor shape.

The visualization uses a standard brain prototype rather than the patient's own anatomy, so spatial relationships between the tumor and patient-specific landmarks (eloquent cortex, motor areas, vessels) are not preserved. This is a significant limitation for any neurosurgical-planning use.

4.3 Atlas-Based Localization Uncertainty

Harvard–Oxford atlas mapping provides approximate neuroanatomical localization that degrades when registration quality is poor. Uncertainty-aware language softens the wording when support is weak, but localization descriptors should be read as orientation aids rather than definitive anatomical assignments, particularly near atlas boundaries or in regions with high inter-subject variability.

4.4 Dataset and Generalizability Constraints

While the project uses publicly available datasets (BraTS 2020 for brain tumor, Medical Segmentation Decathlon Task06_Lung for lung nodule, and Task003_Liver for liver tumor), several generalizability concerns remain:

- **Institutional bias:** The BraTS 2020 dataset, although multi-institutional, primarily represents glioma cases from academic medical centers. Performance on data from community hospitals or low-resource settings with different scanner protocols remains untested.
- **Pathology scope:** The system is currently validated for glioma (brain, BraTS 2020, $n = 5$ fold-0 cases), lung nodule (Task06_Lung, $n = 13$), and liver tumor segmentation (Task003_Liver, $n = 10$). Extension to other pathologies (meningiomas, metastases, vascular malformations) would require additional training data and validation.
- **Population diversity:** The demographic composition of the training datasets may not fully represent global patient populations, potentially introducing bias in segmentation performance across different ethnic groups or age ranges.
- **Temporal limitations:** The system processes single-timepoint studies. Longitudinal tracking of tumor progression or treatment response would require additional pipeline components.

4.5 VR Hardware and Accessibility

The Meta Quest 3, while offering excellent standalone VR capabilities, introduces several practical limitations. The mobile GPU imposes constraints on rendering quality and volume

resolution. The $0.65\times$ sampling-rate multiplier applied on Android reduces the effective ray-marching step count from 512 to roughly 333 and can under-sample structures thinner than the voxel step, which is a concern for millimetric lesions and thin septa. In addition, tricubic interpolation is disabled on Android for an approximately $8\times$ reduction in texture reads per sample; the resulting voxel aliasing is most visible in low-gradient regions such as normal parenchymal transitions. Together these two mobile-only shader choices may cause subtle visual artifacts in regions of gradual density change. Additionally, the VR interaction paradigm, while intuitive for many users, presents accessibility barriers for individuals with vestibular disorders, visual impairments, or limited hand mobility.

The need for a PC workstation for preprocessing limits deployment in resource-constrained clinical environments; a cloud-based alternative would lift that constraint but reintroduce privacy and latency concerns.

Beyond GPU and ergonomic considerations, the Unity-side deployment itself introduces additional, more operational limitations:

- **Sideload-based case transfer:** Each new case must be copied into the headset's persistent data directory via USB sideload (adb push or Meta Quest Developer Hub). The system has no headset-side case browser or PACS/DICOM ingestion layer.
- **Single-scene, single-case architecture:** The VR build ships with a single Unity scene initialized for one volume at a time; switching cases effectively requires leaving and re-entering the application, precluding VR-side playlists and multi-case comparison within one session.
- **No in-VR annotation or measurement:** The interaction model supports sphere-driven slicing plane control and a transfer-function toggle, but no volumetric measurement, bookmarking or on-volume annotation.
- **Narrow transfer-function library:** Only two transfer-function presets are implemented (plugin default and CT-Lung); additional modality- or tissue-specific presets would require code changes.
- **No axial/coronal/sagittal snap on slicing plane:** The slicing plane is driven continuously by the user sphere; there is no preset snap to the standard radiological planes.
- **Controller-only interaction:** All user input is bound to Meta Quest controllers; hand-tracking is not used.

4.6 Evaluation Methodology Gaps

The current evaluation relies primarily on voxel-level segmentation metrics (Dice, Precision, Recall) and system-level quality checks. Several important areas are not yet covered:

- **Clinical utility assessment:** No formal user studies with clinicians or medical students have been conducted to measure the system's impact on spatial comprehension, diagnostic accuracy, or surgical planning outcomes.

- **VR usability evaluation:** System Usability Scale (SUS) scores and task completion time measurements have not been collected.
- **Learning outcome measurement:** Pre-test/post-test studies comparing VR-based learning with traditional methods have not been performed.
- **Long-term retention:** The effects of VR-based medical education on long-term knowledge retention remain unmeasured.

5. Ethical Considerations

5.1 Patient Data Privacy and Anonymization

MedVisVR uses only publicly available, de-identified, IRB-approved research datasets (BraTS 2020, Medical Segmentation Decathlon Task06_Lung, and Task003_Liver), which removes the primary-collection consent problem. Three residual privacy points remain relevant in practice:

- **Offline by construction:** Segmentation, metrics, atlas mapping, the FastAPI service, the local llama.cpp LLM and the VR all run on the workstation and headset without network dependency. No patient image, report or question ever leaves the local environment, which protects PHI by construction and allows deployment in low-connectivity settings.
- **Headset-side residuals:** NIFTI files written to the headset's persistent data directory remain until manually removed. There is no session-end purge, encrypted-at-rest storage or per-case access control at the VR layer. Shared headsets therefore need an explicit deletion / session-wipe protocol. Additionally, the Quest 3 platform layer (Meta account, OS-level updates, device telemetry) sits outside the application sandbox and must be operated in Meta Quest for Business / MDM mode for strict HIPAA/GDPR environments.
- **Visualization identifiability:** Skull-stripping mitigates re-identification via facial reconstruction, but any future extension to non-stripped data must revisit this. Immersive 3D renderings also carry higher recognition potential than 2D slices and should be anonymized appropriately for educational demonstrations.

5.2 Clinical Responsibility and Liability

MedVisVR is explicitly positioned as a research and educational decision-support prototype rather than a clinical diagnostic system. This distinction matters for several reasons:

- **Regulatory status:** The system has not undergone the regulatory approval process required for clinical medical devices (CE marking in Europe, FDA clearance in the United States). Using it for real clinical decisions without that approval would be both ethically and legally problematic.
- **Disclaimer framework:** The report generation pipeline includes mandatory disclaimer sections and safety statements, which is a responsible approach. However, the

effectiveness of these disclaimers depends on users actually reading and understanding them, which cannot be guaranteed in practice.

- **Over-reliance risk:** The professional quality of the VR visualizations and AI-generated reports could create a false sense of clinical reliability. Users may be tempted to treat the system's outputs as validated clinical findings rather than research-grade approximations.

5.3 Algorithmic Bias and Fairness

Deep learning models trained on specific datasets inevitably reflect the characteristics and biases of those datasets. For MedVisVR, the ethical implications include:

- **Demographic representation:** If the training datasets disproportionately represent certain demographics, the model's segmentation accuracy may vary across patient populations. This could lead to disparate quality of visualization and decision support for underrepresented groups.
- **Pathology spectrum:** The system's focus on glioma, lung nodule, and liver tumor segmentation means that other brain pathologies may be inadequately represented or misinterpreted if the system is applied beyond its validated scope.
- **Transparency:** The system's evidence-linked outputs and confidence scoring provide a degree of algorithmic transparency that helps users assess the reliability of individual outputs. However, current transparency is limited to aggregate confidence scores; voxel-level explainability and audit trails remain absent and should be addressed in future work to meet clinical accountability standards.

5.4 Informed Use in Medical Education

When deploying MedVisVR in educational settings, ethical considerations include ensuring that students understand the system's limitations and do not develop uncritical trust in AI-generated medical outputs. The system should be presented as a tool that augments, rather than replaces, traditional medical education methods. Educators should emphasize that the VR visualizations are approximations derived from automated segmentation and should always be validated against clinical expertise.

5.5 Intellectual Property and Licensing Compliance

The project adheres to the specific licensing terms of each dataset (primarily Creative Commons Attribution 4.0 International). The use of open-source tools (nnU-Net, UnityVolumeRendering, llama.cpp) also requires compliance with their respective licenses. The scientific integrity commitment to referencing all original authors and institutions ensures proper attribution and reproducibility.

5.6 VR-Induced Discomfort and Informed Consent

The VR layer introduces an ethical consideration that has no direct analog on the desktop side: head-mounted display use can produce cybersickness, eye strain and disorientation, with a non-negligible incidence in first-time users and in individuals with pre-existing vestibular or ocular

conditions. The system’s performance optimizations (stable 72 Hz refresh rate, Fixed Foveated Rendering, mobile-tuned sampling rate) are designed to reduce this risk at the engineering level, but they do not remove it. In educational deployments, particularly where medical students, trainees or patient participants are asked to use the headset as part of a teaching session or user study, this implies a specific informed-consent obligation: participants should be told in advance that the session involves immersive VR, what the known side effects are, that they may stop at any time without consequence, and that time-limited exposure and rest breaks are available. This consideration should be reflected in any institutional review protocol accompanying the deployment of MedVisVR in a learning context.

6. Scalability Analysis

6.1 Computational Scalability

The MedVisVR pipeline has several computationally heavy stages. Scaling to larger deployments requires addressing the following:

6.1.1 Segmentation Throughput

nnU-Net inference on a single brain MRI case with GPU acceleration takes approximately 1-2 minutes, while CPU inference can exceed 30 minutes. For clinical deployment scenarios requiring rapid turnaround, this inference time may be acceptable with GPU acceleration but is prohibitive on CPU-only systems. Scaling to high-throughput environments would require either dedicated GPU infrastructure or cloud-based batch processing.

6.1.2 VR Rendering Performance

The Meta Quest 3’s mobile GPU (Adreno 740, part of the Snapdragon XR2 Gen 2 SoC) constrains the rendering complexity. The current optimization strategy (a 0.65× sampling-rate multiplier, corresponding to roughly 333 effective ray-marching steps down from 512; tricubic interpolation disabled on Android; Fixed Foveated Rendering; 72 Hz refresh rate) achieves acceptable performance for single-volume visualization. However, simultaneous rendering of multiple volumes, higher-resolution datasets, or additional anatomical overlays would strain the available GPU budget and may require more aggressive level-of-detail strategies or streaming approaches. Three further scalability boundaries follow directly from the present Unity implementation. First, the Quest 3 exposes roughly 8 GB of shared memory with a practical per-application ceiling below that, so a single float-16 volume at 512³ already occupies on the order of 268 MB before counting slice render targets, shader-side atlases and UI; 1024³ volumes or multi-modal overlays (MRI + CT fusion) would exceed the budget without tiling or downsampling. Second, the runtime volume bootstrap is a single-shot builder: it tears the previous sphere down and constructs a new GameObject hierarchy on load, which is adequate for case-at-a-time review but is not a hot-swap or streaming architecture. Third, loading large NIFTI files from the headset’s internal storage introduces a user-visible latency that can reach tens of seconds for volumes around 500 MB; the on-device loading HUD mitigates the user-experience impact but does not remove the underlying I/O bottleneck. Closely related is PC-tethered

rendering via Quest Link / Air Link, which would lift most of these mobile limits at the cost of reintroducing a host-to-headset dependency.

6.2 Multi-Pathology Extension

Expanding MedVisVR beyond brain glioma, lung nodule, and liver tumor segmentation to support additional pathologies is architecturally feasible but requires:

- **Additional training data:** Each new pathology type requires curated, annotated datasets and dedicated model training or fine-tuning.
- **Transfer function design:** New pathology types may require custom transfer functions for optimal VR visualization.
- **AI assistant adaptation:** The structured descriptor layer, question classification, and safety rules would need to be extended for each new clinical domain.
- **Validation studies:** Each pathology extension would require independent validation to ensure segmentation accuracy and visualization quality meet established standards.

6.3 Institutional Deployment

Deploying MedVisVR in clinical or educational institutions presents several scalability challenges:

- **IT infrastructure:** Institutions would need CUDA-capable workstations for preprocessing, Meta Quest 3 headsets for visualization, and potentially network infrastructure for data transfer.
- **Data integration:** Clinical deployment would require DICOM-to-NIfTI conversion pipelines and integration with hospital PACS systems.
- **User training:** Both the desktop application and VR interface require user training. More comprehensive onboarding workflows would be needed.
- **Maintenance and updates:** Keeping segmentation models current, updating VR applications, and maintaining hardware would require dedicated technical support.

7. Comparison with Existing Approaches

Within the existing literature, the contribution of MedVisVR is integrative rather than foundational. The system does not introduce new segmentation architectures or new VR rendering techniques. Its value lies in the end-to-end combination of established methods into a coherent, usable pipeline.

Compared with structured clinical reporting frameworks such as BT-RADS, MedVisVR is less clinically mature but more automated at the case-processing level. Compared with LLM-first report generation approaches, the system is more conservative and more tightly grounded in deterministic case data. Compared with segmentation-only studies, MedVisVR contributes less at

the model-innovation level but significantly more at the level of explainability, downstream usability, and immersive visualization.

The main differentiator is the integrated, fully offline pipeline: automated segmentation, quantitative analysis, atlas-aware localization, grounded reporting and interactive VR visualization, all executed locally without cloud dependencies. Many comparable systems rely on cloud-hosted LLM APIs or remote inference services, which creates patient-privacy friction and makes them unusable in disconnected clinical environments. We are not aware of a publicly documented system that combines these capabilities in a single end-to-end, locally executable workflow covering both standalone VR visualization and desktop-based clinical review.

8. Future Directions

Based on the limitations and scalability analysis presented above, the following directions are identified for future development:

- **Segmentation uncertainty visualization:** Incorporating voxel-level confidence maps into the VR rendering to provide users with intuitive understanding of segmentation reliability.
- **Deformable registration:** Implementing more sophisticated spatial alignment methods to improve anatomical accuracy for surgical planning applications.
- **Clinical validation studies:** Conducting formal user studies with clinicians and medical students to measure the system's impact on spatial comprehension, learning outcomes, and planning accuracy.
- **Multi-pathology support:** Extending the pipeline to support additional neurological and non-neurological pathologies.
- **Longitudinal tracking:** Adding capabilities for temporal comparison of sequential imaging studies to support treatment response monitoring.
- **Multi-user collaboration:** Implementing networked VR sessions for collaborative surgical planning and group medical education.
- **Regulatory pathway:** Initiating the regulatory approval process to transition the system from a research prototype to a clinically deployable medical device.

9. Conclusion

This chapter examined MedVisVR across four areas: experimental results, limitations, ethical considerations and scalability. The system brings together deep-learning segmentation, quantitative analysis, atlas-aware localization, grounded clinical reporting and immersive VR visualization in a single pipeline.

The segmentation results, per-sub-region Dice of 0.80–0.87 for brain, 0.95 parenchyma and 0.71 tumor for liver, and a bimodal 0.65 mean Dice for lung driven by one under-segmentation failure, sit in the range expected from pretrained nnU-Net backbones and are sufficient for educational and research use. The VR stack, with its 4× intensity enhancement, multi-scale boundary blending and interactive slicing, provides intuitive access to complex 3D medical data on affordable standalone hardware.

Significant limitations remain: segmentation-dependent error propagation, rigid-registration accuracy, the absence of clinical validation, and Quest 3 hardware and accessibility constraints. The ethical analysis reinforces the research-prototype positioning, the privacy guarantees (and the platform-layer caveats around them), and the need to address algorithmic bias. The scalability analysis identifies both technical ceilings (GPU budget, storage I/O, single-scene architecture) and extension paths (multi-pathology, institutional deployment, PC-tethered rendering).

Overall, MedVisVR is a working prototype for accessible, immersive medical visualization. Taking it from a research prototype to a clinically validated tool will require further work on validation, regulatory compliance and user-centered design.